

Setting Up GCC and CMake on Windows (MSYS2 + UCRT64)

A quick, copy-and-paste guide to get the C++ toolchain running. No prior technical experience needed.

Time needed: about 15 to 20 minutes (mostly waiting for downloads).

Step 0: Open the MSYS2 download page

Go to: <https://www.msys2.org/>

Step 1: Install MSYS2

1. On the page above, click the big download button for the installer (it looks like `msys2-x86_64-xxxxxxx.exe`).
2. Double-click the downloaded file to run it.
3. Click **Next** through the screens. **Keep the default install folder:** `C:\msys64` (the rest of this guide assumes it).
4. Click **Install**, wait for it to finish, then click **Finish**.

If the installer opens a terminal window at the end, close it. We'll use a specific one in the next step.

Step 2: Update MSYS2 using the UCRT64 terminal

1. Press the **Windows** key, type **MSYS2 UCRT64**, and open it.
Make sure the window says **UCRT64**, not "MSYS" or "MINGW64". The correct one has a yellow-ish icon.
2. Copy this command, paste it into the terminal (right-click to paste), and press **Enter**:

```
bash
```

```
pacman -Syu
```

3. When asked `Proceed with installation? [Y/n]`, type **Y** and press **Enter**.

⚠ The terminal may close itself (this is normal)

During this first update, MSYS2 may update its own core files. When that happens you will see a message such as *"terminate MSYS2 without returning to shell and check for updates again"* and the window may **close on its own** or ask you to close it.

If this happens:

1. **Close the UCRT64 terminal** (if it has not already closed).
2. **Open MSYS2 UCRT64 again** from the Start menu.
3. **Run the same command again:**

```
bash  
  
pacman -Syu
```

4. Press **Y** then **Enter** when asked.

Repeat until you see `there is nothing to do`. That means everything is up to date.

Step 3: Install GCC, CMake and Make

In the **UCRT64** terminal, paste this single command and press **Enter**:

```
bash  
  
pacman -S mingw-w64-ucrt-x86_64-gcc mingw-w64-ucrt-x86_64-cmake mingw-w64-ucrt-
```

- If asked to choose from a list or `Proceed with installation? [Y/n]`, just press **Enter** (accept the default) or type **Y** then **Enter**.
- Wait until it finishes and you get your prompt back.

The tools are now installed in:

```
C:\msys64\ucrt64\bin
```

You will add this folder to Windows in the next step so that PowerShell, VS Code, etc. can find the tools.

Step 4: Add the `bin` folder to your Windows PATH

1. Press the **Windows** key, type **environment variables**, and click **Edit the system environment variables**. (Or: "Edit environment variables for your account".)
2. In the window that opens, click **Environment Variables...**
3. Choose **one** of the following:

Option	Where	When to use
User variables (top box)	Only affects your account	Recommended. No admin rights needed.
System variables (bottom box)	Affects all accounts on the PC	Only if you have admin rights and want it for everyone.

4. In the chosen box, select the variable named **Path** and click **Edit...**
5. Click **New** and type exactly:

```
C:\msys64\ucrt64\bin
```

6. Click **OK** on every open window to save (Edit → Environment Variables → System Properties).

Tip: Don't delete or change any existing entries in the list. Only add the new one.

Step 5: Verify the installation in PowerShell

Important: Open a **new** PowerShell window **after** saving the PATH change. Windows that were already open will not see the update.

A laptop restart is normally NOT required. Closing and reopening PowerShell (or VS Code / your IDE) is enough. Only restart if the commands below are still "not recognized" after reopening PowerShell.

1. Press the **Windows** key, type **PowerShell**, and open it.
2. Run each command below, one at a time:

```
powershell
```

```
gcc --version
g++ --version
cmake --version
mingw32-make --version
```

3. Each command should print a version number, for example:

```
gcc.exe (Rev..., Built by MSYS2 project) 15.x.x
cmake version 4.x.x
```

If all four print a version, **you're done!** 

Troubleshooting

Problem	Fix
'gcc' is not recognized... or 'cmake' is not recognized...	Close all PowerShell windows and open a new one. If it still fails, restart the laptop once.
Still not recognized after restart	Re-check Step 4: the path must be exactly <code>C:\msys64\ucrt64\bin</code> (no typos, no extra spaces).
Opened the wrong MSYS2 terminal	Close it and open MSYS2 UCRT64 specifically.
<code>pacman</code> errors about locked database or failed downloads	Close all MSYS2 windows, reopen UCRT64 , and run <code>pacman -Syu</code> again.
You installed MSYS2 somewhere other than <code>C:\msys64</code>	Use your actual folder in Step 4, e.g. <code>D:\msys64\ucrt64\bin</code> .

Quick checklist

- ☐ MSYS2 installed in `C:\msys64`
- ☐ `pacman -Syu` repeated until it says *nothing to do*
- ☐ GCC, CMake and Make installed with `pacman -S ...`

☐ `C:\msys64\ucrt64\bin` added to **Path**

☐ New PowerShell shows versions for `gcc`, `g++`, `cmake`, `mingw32-make`